

Independent Replication Report:
*An Application of Computational Fluid Dynamics to
Optimize Municipal Sewage Networks; A Case of Tororo
Municipality, Eastern Uganda*
(Esemu et al., 2020, JAM v18)

OpenClaw main-agent replication (subagent 147de903)
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Abstract

Independent replication of Esemu, Masanja, Nampala, Lwanyaga, Awichi & Semwogerere (2020), which combines an analytical Manning-equation table of minimum sewer slopes with an OpenFOAM `interFoam` VOF + $k-\varepsilon$ simulation of a 2-D 20 m $\times$ 0.5 m pipe. **Verdict: PARTIAL.** Table 1 is cleanly quantitatively replicated (back-solved Manning $n = 0.0129 \pm 0.0009$, standard concrete-sewer roughness). The CFD figures are qualitatively reproduced via an independently-executed `interFoam` v1906 run on uicgpu; the paper reports no numeric CFD tables, so full quantitative CFD agreement is not verifiable *by the paper’s own reporting*. The municipal-connections uplift claim ($535 \rightarrow 1200 \Rightarrow +80\%$) is out of scope (requires utility records). A wave-brief metadata mismatch (brief said “Al Manazlah, Saudi Arabia”; the DOI actually resolves to the Tororo, Uganda paper) is flagged and we replicated the actual paper.

1 Paper summary

Small (10-page) applied CFD / mathematics paper motivated by Tororo Municipality, Uganda’s under-capacity combined sewer system. Three intertwined claims:

1. An analytical Table 1 of minimum longitudinal slopes for $D \in \{150, 200, 250, 300, 375, 450, 525, 600\}$ mm at self-cleansing velocity $v_0 = 0.6$ m/s.
2. A CFD demonstration using OpenFOAM `interFoam` (VOF two-phase + $k-\varepsilon$) on a 2-D 20 m $\times$ 0.5 m pipe (Figures 2–10). No numeric tables from the CFD are given, only field plots.
3. A municipal recommendation: expand connections from 535 (31.2%) to ≥ 1200 (70%) to raise sewage delivery to treatment by $\sim 80\%$.

2 Claims table

ID	Claim	Type	Testable?	Tested / Result
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C1	Table 1 slopes $S(D)$ are correct minimum-slope-for-self-cleansing values for $D \in \{150 \dots 600\}$ mm	analytical	yes	tested $\rightarrow$	REPLICATED
C2	interFoam on 2-D 20 m $\times$ 0.5 m pipe produces the qualitative Figs 2,3,7,8	numerical	partly (no numeric ground truth)	tested $\rightarrow$	QUALITATIVE MATCH
C3	Centerline velocity develops along the pipe (Fig 9)	qualitative	yes	tested $\rightarrow$	QUALITATIVELY REPLICATED
C4	Expand connections 535 $\rightarrow$ 1200 $\Rightarrow$ +80% delivery	municipal/statistical	is not utility records	not tested (out of scope)	
C5	UPVC/HDPE pipes are preferred over metallic/concrete (lower roughness, 30 yr lifetime)	literature	partly	consistent with tables; internal inconsistency with Table 1 flagged	

3 Method

3.1 PDF fetch & extraction

Crossref $\rightarrow$ publisher $\rightarrow$ citation_pdf_url at <https://rajpub.com/index.php/jam/article/download/8345/7894> $\rightarrow$ paper.pdf (10 pp, 406 604 B, PDF v1.7). Text via pdftotext -layout (581 lines); duplicated as extraction/marker.md and extraction/nougat.mmd (central Marker/Nougat manifests not queried for this new PDE-set paper).

3.2 C1 — Manning-equation replication

Manning’s equation for a full-flowing circular pipe:

$$v = \frac{1}{n} R^{2/3} S^{1/2}, \quad R = D/4.$$

Solve for S at $v = v_0 = 0.6$ m/s:

$$S = \left(\frac{v_0 n}{(D/4)^{2/3}} \right)^2.$$

Back-solve n per Table 1 row: $n = \sqrt{S} (D/4)^{2/3} / v_0$. Grid-search a best-fit single-material n over all 8 rows.

3.3 C2, C3 — OpenFOAM interFoam replication

Host / software. uicgpu (Ubuntu 20.04); OpenFOAM v1906 (libopenfoam/focal), source bashrc at /usr/share/openfoam/etc/bashrc; solvers blockMesh and interFoam on /usr/bin.

Geometry. 2-D rectangular domain $L \times H \times \delta = 20 \times 0.5 \times 0.01$ m, one cell thick in z with empty front/back patches $\Rightarrow$ true 2-D. Mesh $400 \times 40 \times 1 = 16\,000$ hex cells (cell size $0.05 \times 0.0125 \times 0.01$ m).

Solver. interFoam (VOF two-phase), RAS $k-\varepsilon$ turbulence with kqRWallFunction, epsilonWallFunction, nutkWallFunction. Water $\rho = 1000$ kg/m³, $\nu = 10^{-6}$ m²/s; air $\rho = 1$, $\nu = 1.48 \times 10^{-5}$; $\sigma = 0.07$ N/m; $\mathbf{g} = (0, -9.81, 0)$.

BCs. Inlet: $\mathbf{U} = (1.0, 0, 0)$ m/s, $\alpha_{\text{water}} = 1$; outlet: `inletOutlet U, totalPressure` $p_0 = 0$; walls: no-slip; front/back: `empty`. Turbulence: $I = 0.05 \Rightarrow k = 0.00375 \text{ m}^2/\text{s}^2$, $\varepsilon = C_\mu^{3/4} k^{3/2}/l$ with $l = 0.07D = 0.035 \text{ m} \Rightarrow \varepsilon = 0.00097 \text{ m}^2/\text{s}^3$.

Numerics. Euler ddt; linearUpwind for $\mathbf{U}$; vanLeer for α ; upwind for k, ε ; Gauss linear for viscous. PIMPLE: 1 outer, 3 correctors. Adjustive dt, $\text{Co}_{\text{max}} = 1$, $\text{Co}_{\alpha, \text{max}} = 1$. $t_{\text{end}} = 5 \text{ s}$, write every 0.5 s, wall-clock $\approx 80 \text{ s}$.

Post-processing. `postProcess -func sampleDict -latestTime: centerline (200 pts)` and two cross-sections at $x = 5, 15 \text{ m}$ (50 pts each), fields $\mathbf{U}, p, p_{\text{rgh}}, \alpha_{\text{water}}$.

4 Results vs paper

4.1 C1 — Table 1 quantitative reproduction

D (mm)	S paper	n back-solved	S predicted ($n = 0.0129$)	rel-err
150	0.00430	0.01224	0.00478	+11%
200	0.00330	0.01299	0.00325	−1.5%
250	0.00250	0.01312	0.00242	−3.2%
300	0.00190	0.01292	0.00189	−0.5%
375	0.00140	0.01287	0.00141	+0.7%
450	0.00110	0.01288	0.00110	0%
525	0.00092	0.01306	0.00089	−3.3%
600	0.00077	0.01306	0.00074	−3.9%

Back-solved n range: $[0.01224, 0.01312]$, mean 0.01289. This unambiguously matches standard concrete/vitrified-clay sewer roughness ($n \sim 0.013$). The 150 mm row is a mild outlier ($n = 0.0122$, +11% slope over strict-Manning fit). **C1 clean quantitative REPLICATED.** Internal inconsistency: paper’s §4.1 recommends UPVC/HDPE ($n \sim 0.009$), for which Table 1 slopes would be $\sim 2.1\times$ smaller — flagged as Open Question Q1.

4.2 C2, C3 — CFD qualitative reproduction

Quantity	Value
U_x at inlet (first cell, $x = 0.01 \text{ m}$)	1.001 m/s
U_x max along centerline	2.172 m/s at $x = 1.32 \text{ m}$
U_x at outlet (last cell)	1.058 m/s
U_x mean over last 5 m	0.855 m/s
p (kinematic) at inlet	1503 m^2/s^2
p (kinematic) at outlet	−2.5 m^2/s^2
α_{water} mean over centerline	0.047 (front $\approx 5 \text{ m}$ of 20 m at $t = 5 \text{ s}$)
Solver wall-clock	80.4 s

Fig. 9 (paper): unlabelled-axis plot showing velocity development along the channel. Our run reproduces the qualitative shape — inlet $\rightarrow$ entrance overshoot $\rightarrow$ developed downstream regime. See `report/evidence/openfoam_case1/cfd_replication_figures.png`.

Figs 2, 3, 7, 8 (paper): unlabelled colour-field plots. Our centerline + cross-section extractions are consistent with the pressurized full-pipe-water behaviour those fields depict.

4.3 C4 — municipal connections claim

Not tested. Requires NWSC / Busitema municipal records not indexed publicly.

5 Verdict + justification

PARTIAL. C1 is fully quantitatively replicated with a clean Manning derivation (back-solved $n = 0.013 =$ standard concrete sewer). C2/C3 are qualitatively replicated via an actually-executed `interFoam` v1906 run; the paper reports no numeric CFD tables, so quantitative CFD agreement is unreachable by the paper’s own reporting. C4 (municipal claim) not tested.

The replication surfaced two substantive concerns the paper does not address: material-inconsistency between Table 1 (concrete $n = 0.013$) and §4.1’s UPVC/HDPE recommendation, and the 2-D-vs-3-D hydraulic-radius mismatch that makes the paper’s own CFD unable to validate its own Table 1.

6 Open Questions

Q1. Given Table 1’s back-solved $n = 0.013$ (concrete) but §4.1’s UPVC/HDPE recommendation ($n \approx 0.009$), which set of minimum slopes should Tororo actually adopt? At $n = 0.009$ the minima are $\sim 2.1\times$ steeper — a materially different design table.

Next steps: rebuild Table 1 at $n \in \{0.009, 0.011, 0.013\}$; publish a labeled-by-material design table.

Q2. Is the $\sim 2\times$ entrance-region velocity overshoot in the reproduced Fig. 9 physical, or a fixedValue-U inlet + dry-domain start-up artefact?

Next steps: rerun with `flowRateInletVelocity` BC and a pre-filled water domain; compare developed-regime velocities.

Q3. How much does the paper’s 2-D rectangular representation of a 3-D circular pipe shift the developed velocity? For $D = 0.5$ m the 2-D hydraulic radius ($R = h/2 = 0.25$) is $2\times$ the true circular value ($R = D/4 = 0.125$), so Manning velocity differs by $2^{2/3} \approx 1.59\times$. The paper’s own CFD therefore cannot validate its own Table 1.

Next steps: build a 3-D axisymmetric `wedge`-patch case; compare developed velocity at $x = 15$ m.

Q4. No mesh-convergence study is reported. What is the interface-diffusion error on α_{water} and the developed-velocity Richardson error for $400 \times 40 \rightarrow 800 \times 80 \rightarrow 1600 \times 160$?

Next steps: three-level GCI study; publish converged values.

Q5. Does a full 1-D network-scale (SWMM/EPANET) solve on the real Tororo topology, driven by Table 1 slopes, actually deliver the paper’s asserted 80% collection uplift under peak-wet-weather flow, or is that a linear extrapolation that breaks?

Next steps: build SWMM model of Tororo sewer network with 535-baseline and 1200-expansion; force with dry-weather and 1-in-2-year storm; report delivered fraction and CSO overflow count.

Bibliography note

Paper’s own bibliography (references 1–22) preserved in `extraction/marker.md`. Manning-equation reference: Chow, V.T., *Open-Channel Hydraulics*, McGraw-Hill, 1959.